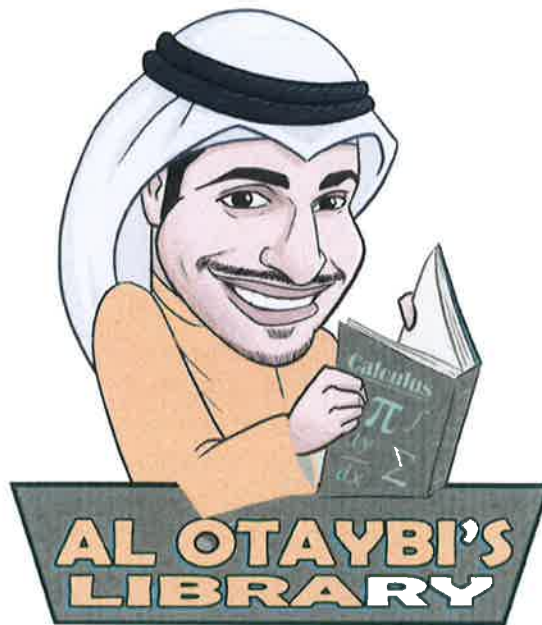


Kuwait University

2021/2022

Spring course



Calculus B

Second

Section : 11.1 (Part d/d)

11.1 Sequences.

(8 d)

* Trick *

Convergent or Divergent ?

$$a_n = \left(\frac{2n+1}{3n+2} \right)^n$$

$$\lim_{n \rightarrow \infty} \left(\frac{2n+1}{3n+2} \right)^n$$

ما ينفع أبني الحل
على
للاختلاف مقامات
n

$$\lim_{n \rightarrow \infty} \left(\frac{\cancel{n} [2 + \frac{1}{n}]}{\cancel{n} [3 + \frac{2}{n}]} \right)^n$$

$$= \left(\frac{2+0}{3+0} \right)^\infty = \left(\frac{2}{3} \right)^\infty = 0$$

$\therefore a_n$ is convergent.

$$a_n = \begin{cases} \frac{n^2}{2n+1} \sin\left(\frac{\pi}{n}\right) & , n \text{ is even} \\ n^2 \ln\left(1 + \frac{2}{n^2}\right) & , n \text{ is odd} \end{cases}$$

when n is even

$$\lim_{n \rightarrow \infty} \frac{n^2}{2n+1} \sin\left(\frac{\pi}{n}\right)$$

$\sin 0$

$$\lim_{n \rightarrow \infty} \frac{n^2}{2n+1} \frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}} \frac{\pi}{n}$$

$$\lim_{n \rightarrow \infty} \frac{\cancel{n^2}}{n(2+\frac{1}{n})} \cancel{\frac{\pi}{n}} \frac{\sin\left(\frac{\pi}{n}\right)}{\frac{\pi}{n}}$$

$$= \frac{\pi}{2+0} (1) = \frac{\pi}{2}$$

when n is odd

$$\lim_{n \rightarrow \infty} n^2 \ln\left(1 + \frac{2}{n^2}\right)$$

$\infty \ln(1)$
 $\infty \quad 0$

$$\lim_{n \rightarrow \infty} \ln\left(1 + \frac{2}{n^2}\right)^{n^2}$$

$$= \ln(e^2)$$

$$= 2$$

$$\therefore \lim_{n \rightarrow \infty} a_n \text{ (when } n \text{ is even)}$$

$$\neq \lim_{n \rightarrow \infty} a_n \text{ (when } n \text{ is odd)}$$

$$\Rightarrow \therefore \lim_{n \rightarrow \infty} a_n \text{ DNE}$$

$$\Rightarrow a_n \text{ is Divergent}$$

$$a_n = \frac{3^{n+1} + n^2}{(-3)^n + 2^n}$$

$$a_n = \begin{cases} \frac{3^{n+1} + n^2}{-3^n + 2^n} \\ \frac{3^{n+1} + n^2}{3^n + 2^n} \end{cases}$$

when n is odd

when n is even

when n is odd

$$\lim_{n \rightarrow \infty} \frac{3^n \cdot 3 + n^2}{-3^n + 2^n} \div 3^n$$

$$\lim_{n \rightarrow \infty} \frac{3 + \frac{n^2}{3^n}}{-1 + (\frac{2}{3})^n}$$

$$\lim_{x \rightarrow \infty} \frac{x^2}{3^x}$$

$$\text{L'Hop} \lim_{x \rightarrow \infty} \frac{2x}{3^x \ln 3}$$

$$\text{L'Hop} \lim_{x \rightarrow \infty} \frac{2}{3^x (\ln 3)^2}$$

$$= \frac{2}{\infty} = 0$$

$$= \frac{3+0}{-1+0} = -3$$

when n is even

$$\lim_{n \rightarrow \infty} \frac{3^n \cdot 3 + n^2}{3^n + 2^n} \div 3^n$$

$$\lim_{n \rightarrow \infty} \frac{3 + \frac{n^2}{3^n}}{1 + (\frac{2}{3})^n}$$

$$= \frac{3+0}{1+0}$$

$$= 3$$

$$\lim_{n \rightarrow \infty} a_n$$

when n is odd

$$\neq \lim_{n \rightarrow \infty} a_n$$

when n is even

$$\lim_{n \rightarrow \infty} a_n \text{ DNE}$$

$\lim_{n \rightarrow \infty} a_n$ is Diverge

Show that the sequence defined by

$$a_1 = 1$$

$$a_{n+1} = 5 + a_n$$

is bounded below

Let $a_1 = 1$

$$a_n \geq 5 \quad \text{(Now we have to prove that } a_{n+1} \geq 5 \text{)}$$

$$+ 5$$

$$5 + a_n \geq 10$$

$$\Downarrow$$

$$a_{n+1} \geq 10 \geq 5$$

$$\Rightarrow \forall a_{n+1} \geq 5$$

$\therefore$ the sequence is bounded below

Show that the sequence defined by

$$a_1 = 1$$

$$a_{n+1} = 5 - \frac{1}{a_n} \text{ is bounded above.}$$

$$a_1 = 1, \quad a_n < 5$$

(Now we have to prove that

$$a_{n+1} < 5)$$

* -1

$$-a_n > -5$$

$$-\frac{1}{a_n} < -\frac{1}{5}$$

+5

$$5 - \frac{1}{a_n} < 5 - \frac{1}{5}$$

$$5 - \frac{1}{a_n} < \frac{25}{5} - \frac{1}{5}$$

$$5 - \frac{1}{a_n} < \frac{24}{5}$$

↓

$$a_{n+1} < \frac{24}{5} < 5$$

$$\Rightarrow \therefore a_{n+1} < 5$$

Show that the sequence defined by

$$a_1 = 1$$

$a_{n+1} = 5 - \frac{1}{a_n}$ is increasing.

$$n=1 \rightarrow a_2 = 5 - \frac{1}{a_1} = 5 - \frac{1}{1} = 4 \rightarrow a_2 = 4$$

$$n=2 \rightarrow a_3 = 5 - \frac{1}{a_2} = 5 - \frac{1}{4} = \frac{20}{4} - \frac{1}{4} = \frac{19}{4}$$

$$\rightarrow a_3 = \frac{19}{4}$$

$$n=3 \rightarrow a_4 = 5 - \frac{1}{a_3} = 5 - \frac{1}{\frac{19}{4}} = 5 - \frac{4}{19} = \frac{95-4}{19} = \frac{91}{19}$$

$$\rightarrow a_4 = \frac{91}{19}$$

$$\Rightarrow a_n = \{1, 4, \frac{19}{4}, \frac{91}{19}, \dots\}$$

$$a_1 < a_2 < a_3 < a_4 < \dots < a_n < a_{n+1}$$

Let $a_{n+1} > a_n$ (Now we have to prove that

* -1

$$-a_{n+1} < -a_n$$

$$-\frac{1}{a_{n+1}} > -\frac{1}{a_n}$$

+5

$$5 - \frac{1}{a_{n+1}} > 5 - \frac{1}{a_n}$$

$$a_{n+2} > a_{n+1}$$

$\therefore$ the sequence is increasing.

Show that the sequence defined by

$$a_1 = 2 \quad a_{n+1} = \frac{1}{3-a_n}$$

is decreasing.

$$a_1 = 2$$

$$n=1 \rightarrow a_2 = \frac{1}{3-a_1} = \frac{1}{3-2} = \frac{1}{1} = 1$$

$$n=2 \rightarrow a_3 = \frac{1}{3-a_2} = \frac{1}{3-1} = \frac{1}{2}$$

$$n=3 \rightarrow a_4 = \frac{1}{3-a_3} = \frac{1}{3-\frac{1}{2}} = \frac{1}{\frac{6}{2}-\frac{1}{2}} = \frac{1}{\frac{5}{2}} = \frac{2}{5}$$

$$\Rightarrow a_n = \left\{ 2, 1, \frac{1}{2}, \frac{2}{5}, \dots \right\}$$

$$a_1 > a_2 > a_3 > a_4 > \dots > a_n > a_{n+1}$$

$$\text{Let } a_{n+1} < a_n$$

(Now we have to prove that

$$a_{n+2} < a_{n+1}$$

$$*-1$$

$$-a_{n+1} > -a_n$$

$$+3$$

$$3-a_{n+1} > 3-a_n$$

$$\frac{1}{3-a_{n+1}} < \frac{1}{3-a_n}$$

$$a_{n+2} < a_{n+1}$$

$\therefore$ the sequence is decreasing

① Show that the sequence defined by

$$a_1 = 1 \quad a_{n+1} = 3 - \frac{1}{a_n}$$

is increasing and bounded above.

② Deduce that $\{a_n\}$ is convergent and find its limit.

$$a_1 = 1$$

$$n=1 \rightarrow a_2 = 3 - \frac{1}{a_1} = 3 - \frac{1}{1} = 2 \rightarrow a_2 = 2$$

$$n=2 \rightarrow a_3 = 3 - \frac{1}{a_2} = 3 - \frac{1}{2} = \frac{5}{2} \rightarrow a_3 = \frac{5}{2}$$

$$n=3 \rightarrow a_4 = 3 - \frac{1}{a_3} = 3 - \frac{1}{\frac{5}{2}} = 3 - \frac{2}{5} = \frac{13}{5}$$

$$\Rightarrow \therefore a_n = \left\{ 1, 2, \frac{5}{2}, \frac{13}{5}, \dots \right\} \rightarrow a_4 = \frac{13}{5}$$

$$a_1 < a_2 < a_3 < a_4 \dots$$

$$\text{Let } a_n < a_{n+1}$$

(Now we have to prove that

$$a_{n+1} < a_{n+2})$$

$$-a_n > -a_{n+1}$$

$$-\frac{1}{a_n} < -\frac{1}{a_{n+1}}$$

$$3 - \frac{1}{a_n} < 3 - \frac{1}{a_{n+1}}$$

$$a_{n+1} < a_{n+2}$$

① $\Rightarrow$ the sequence is increasing.

$$a_n < 3$$

(Now we have to prove that $a_{n+1} < 3$)

* -1

$$-a_n > -3$$

$$-\frac{1}{a_n} < -\frac{1}{3}$$

+3

$$3 - \frac{1}{a_n} < 3 - \frac{1}{3}$$

↓

$$a_{n+1} < \frac{8}{3} < 3$$

$$a_{n+1} < 3$$

→ ∴ the sequence is bounded above

From (1) (2) ⇒ ∴ a_n is convergent.
(a_n is increasing and bounded above)

∴ a_n is convergent

$$\lim_{n \rightarrow \infty} a_n = L$$

$$\therefore \lim_{n \rightarrow \infty} a_{n+1} = L$$

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \left[3 - \frac{1}{a_n} \right]$$

$$L = 3 - \frac{1}{L}$$

$$L = \frac{3L-1}{L}$$

$$L^2 = 3L - 1$$

$$L^2 - 3L + 1 = 0$$

$$L = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\begin{aligned} a &= 1 \\ b &= -3 \\ c &= 1 \end{aligned}$$

$$L = \frac{3 \pm \sqrt{9 - 4(1)(1)}}{2(1)} = \frac{3 \pm \sqrt{5}}{2} \Rightarrow \therefore L = \frac{3 + \sqrt{5}}{2}$$

* Trick *

① Show that the sequence defined by

$$a_1 = 2 \quad a_{n+1} = \frac{1}{3-a_n}$$

satisfies $0 < a_n < 2$ is decreasing

② Show that the sequence is convergent and find its limit

$$a_1 = 2$$

$$n=1 \rightarrow a_2 = \frac{1}{3-a_1} = \frac{1}{3-2} = 1 \rightarrow a_2 = 1$$

$$n=2 \rightarrow a_3 = \frac{1}{3-a_2} = \frac{1}{3-1} = \frac{1}{2} \rightarrow a_3 = \frac{1}{2}$$

$$n=3 \rightarrow a_4 = \frac{1}{3-a_3} = \frac{1}{3-\frac{1}{2}} = \frac{1}{\frac{5}{2}} = \frac{2}{5} \rightarrow a_4 = \frac{2}{5}$$

$$\Rightarrow a_n = \left[2, 1, \frac{1}{2}, \frac{2}{5}, \dots \right]$$

$$a_1 > a_2 > a_3 > a_4 \dots > a_n > a_{n+1}$$

Let $a_{n+1} < a_n$

(Now we have to prove that

$* -1$

$+3$

$$-a_{n+1} > -a_n$$

$$3-a_{n+1} > 3-a_n$$

$$\frac{1}{3-a_{n+1}} < \frac{1}{3-a_n}$$

$$a_{n+2} < a_{n+1}$$

① $\therefore$ the sequence is decreasing

$$0 < a_n$$

(Now we have to prove

$$0 < a_{n+1}$$

* -

$$0 > 3 - a_n$$

+ 3

$$3 > 3 - a_n$$

$$\frac{1}{3} < \frac{1}{3 - a_n}$$

$$0 < \frac{1}{3} < a_{n+1}$$

$$0 < a_{n+1}$$

→ ∴ The sequence is bounded below

⇒ From ① ② ⇒ a_n is convergent
(a_n is decreasing and bounded below)

$$\because a_n \text{ is convergent} \begin{cases} \therefore \lim_{n \rightarrow \infty} a_n = L \\ \therefore \lim_{n \rightarrow \infty} a_{n+1} = L \end{cases}$$

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{3 - a_n}$$

$$L = \frac{1}{3 - L}$$

$$L(3 - L) = 1$$

$$3L - L^2 = 1$$

$$L^2 - 3L + 1 = 0$$

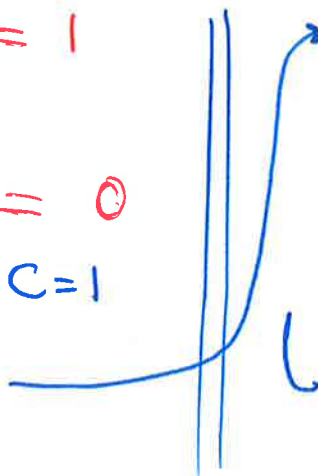
$$a = 1 \quad b = -3 \quad c = 1$$

$$L = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$L = \frac{3 \pm \sqrt{9 - 4(1)(1)}}{2(1)}$$

$$= \frac{3 \pm \sqrt{5}}{2}$$

$$L = \frac{3 - \sqrt{5}}{2} \quad \#$$



Determine whether the following sequence is convergent or divergent

$$a_1 = 2, \quad a_{n+1} = \frac{a_n}{1+a_n}$$

$$a_1 = 2$$

$$n=1 \rightarrow a_2 = \frac{a_1}{1+a_1} = \frac{2}{1+2} = \frac{2}{3} \rightarrow a_2 = \frac{2}{3}$$

$$n=2 \rightarrow a_3 = \frac{a_2}{1+a_2} = \frac{\frac{2}{3}}{1+\frac{2}{3}} = \frac{\frac{2}{3}}{\frac{3+2}{3}} = \frac{\frac{2}{3}}{\frac{5}{3}} \rightarrow a_3 = \frac{2}{5}$$

$$n=3 \rightarrow a_4 = \frac{a_3}{1+a_3} = \frac{\frac{2}{5}}{1+\frac{2}{5}} = \frac{\frac{2}{5}}{\frac{5+2}{5}} = \frac{\frac{2}{5}}{\frac{7}{5}} \rightarrow a_4 = \frac{2}{7}$$

$$n=4 \rightarrow a_5 = \frac{a_4}{1+a_4} = \frac{\frac{2}{7}}{1+\frac{2}{7}} = \frac{\frac{2}{7}}{\frac{7+2}{7}} = \frac{2}{9} \rightarrow a_5 = \frac{2}{9}$$

$$a_n = \left\{ \underset{\textcircled{1}}{2}, \underset{\textcircled{2}}{\frac{2}{3}}, \underset{\textcircled{3}}{\frac{2}{5}}, \underset{\textcircled{4}}{\frac{2}{7}}, \underset{\textcircled{5}}{\frac{2}{9}}, \dots \right\}$$

$$a_n = \frac{2}{2n-1}$$

$$\lim_{n \rightarrow \infty} \frac{2}{2n-1} = \frac{2}{\infty} = 0$$

$\therefore a_n$ is convergent to 0